Name: ___________________________ Class: ____________ Date: ____________ **ID: A**

Chemistry Chapter 17 Practice Quiz 4 — Ksp & Solution Equilibrium

Show ALL work on every calculation. Write what you know → formula → plug in → evaluate.

Multiple Choice

Identify the choice that best completes the statement or answers the question.

___ 1. The solubility product constant (Ksp) is

a. the equilibrium constant for dissolving a solute into a saturated solution
b. the rate at which a solid dissolves in water
c. the concentration of a saturated solution in mol/L
d. the same as Keq for all ionic reactions

___ 2. For the dissolution $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$, the Ksp expression is

a. Ksp = [AgCl]
b. Ksp = $[Ag^+][Cl^-]$ / [AgCl]
c. Ksp = $[Ag^+][Cl^-]$
d. Ksp = $[Ag^+]^2[Cl^-]^2$

___ 3. If the ion concentration product of a solution is GREATER than the Ksp, the solution is

a. unsaturated
b. saturated
c. supersaturated
d. dilute

___ 4. The common-ion effect states that

a. adding a common ion increases the solubility of a salt
b. a common ion from two solutions must be considered in solubility calculations
c. two salts with a common ion will both dissolve completely
d. common ions only affect precipitation reactions

___ 5. For $Cu(OH)_2$ dissolving: $Cu(OH)_2 \rightleftharpoons Cu^{2+} + 2OH^-$. If solubility = S, then Ksp equals

a. $S \times 2S$
b. $S \times S^2$
c. $4S^3$
d. S^2

True / False

Indicate whether the statement is true or false.

___ 6. A Ksp > 1 means the salt is very soluble (large numbers of ions in solution).

___ 7. A salt with a very small Ksp value is considered highly soluble.

___ 8. Adding a second salt with a common ion to a solution can cause precipitation.

___ 9. If the ion concentration product equals Ksp, the solution is saturated.

___ 10. Ksp values can be used to predict whether a precipitate will form when two solutions are mixed.

___ 11. The Ksp expression for a solid dissolving includes the concentration of the solid.

___ 12. For AgCl, $[Ag^+]$ always equals $[Cl^-]$ when AgCl dissolves in pure water.

___ 13. A supersaturated solution has an ion product greater than its Ksp.

___ 14. Ksp can be used to determine the solubility of a salt in mol/L.

___ 15. The common-ion effect increases the solubility of a salt.

Calculations — Show All Work

16. Given that the solubility of $Cu(OH)_2$ is 1.76×10^{-7} M, calculate the Ksp. $Cu(OH)_2 \rightleftharpoons Cu^{2+}(aq) + 2OH^-(aq)$ (Example 17-6 from your textbook)

17. The Ksp of AgCl is 1.8×10^{-10}. Calculate the solubility of AgCl in mol/L. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$ (Example 17-7 from your textbook)

18. A $Zn(OH)_2$ solution has concentration 2.5×10^{-6} M. Ksp for $Zn(OH)_2 = 3.0 \times 10^{-17}$. $Zn(OH)_2 \rightleftharpoons Zn^{2+}(aq) + 2OH^-(aq)$ Calculate the ion concentration product and determine whether the solution is unsaturated, saturated, or supersaturated. (Example 17-8 from your textbook)

Essay

Answer TWO of the following. You may answer all 3 for possible bonus.

19. Define the solubility product constant (Ksp). Explain how Ksp relates to whether a substance is soluble or insoluble. How can Ksp values be used to predict whether a precipitate will form when two solutions are mixed?

20. Explain how to determine whether a solution is unsaturated, saturated, or supersaturated using the Ksp. What does each condition mean physically? What happens at the particle level in each case?

21. Explain the common-ion effect. What is a common ion? How does the presence of a common ion affect the solubility of a second salt? Give a specific example using ions from the PPT.